Exercise Sheet 02 - Discrete Random Variables

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Quick Reference for Well-Known Distributions

When working with common families of discrete random variables, we use specific formulas derived from their parameters rather than calculating them step-by-step from the definition, as we have seen in class.

Distribution	Parameters	$P(X = x)$	$E[X]$	$Var(X)$	$SD(X)$
Bernoulli	p (prob. of success)	$p^x(1-p)^{1-x}$, for $x \in \{0, 1\}$	p	$p(1-p)$	$\sqrt{p(1-p)}$
Binomial	n (trials), p (prob. of success)	$\binom{n}{x}p^x(1-p)^{n-x}$, for $x \in \{0, 1, \dots, n\}$	np	$np(1-p)$	$\sqrt{np(1-p)}$
Poisson	λ (average rate)	$\frac{\lambda^x e^{-\lambda}}{x!}$, for $x \in \{0, 1, 2, \dots\}$	λ	λ	$\sqrt{\lambda}$
Geometric	p (prob. of success)	$(1-p)^{x-1}p$, for $x \in \{1, 2, 3, \dots\}$ *(1st success on trial x)*	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{\sqrt{1-p}}{p}$
Negative Binomial	r (target successes), p	$\binom{x-1}{r-1}p^r(1-p)^{x-r}$, for $x \in \{r, r+1, \dots\}$ *(r-th success on trial x)*	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\frac{\sqrt{r(1-p)}}{p}$

*Note: The Geometric and Negative Binomial formulas above are written for the standard definition counting the total number of **trials** (x). If counting only the number of **failures** before the success, the formulas shift slightly.*

Exercises

Exercise 1

A discrete random variable X has the following probability distribution:

x	$P(X = x)$
1	0.1
2	0.3
3	0.4
4	0.2

Questions:

a) Verify that this is a valid probability distribution.
b) Calculate the expected value (mean) of X, denoted as $E[X]$.
c) Calculate the variance, $Var(X)$, using the computational formula: $Var(X) = E[X^2] - (E[X])^2$.
d) Find the standard deviation, $SD(X)$.

Solution To verify that this is indeed a probability distribution, all probability values must be non-negative (which holds) and sum to 1:

$$\sum_{x=1}^{4} P(X = x) = 0.1 + 0.3 + 0.4 + 0.2 = 1.0.$$

So, this is indeed a probability distribution.

For the mean value, we have the formula:

$$E[X] = \sum_{x=1}^{4} xP(X = x) = 1 \cdot 0.1 + 2 \cdot 0.3 + 3 \cdot 0.4 + 4 \cdot 0.2 = 2.7.$$

We can also compute it using R:

```
x <- 1:4
pmf <- c(0.1, 0.3, 0.4, 0.2) # Creates a column, i.e., a vector.
mean_value <- sum(x * pmf)
# x   = [1, 2, 3, 4]
# pmf = [0.1, 0.3, 0.4, 0.2]
# x * pmf = [0.1, 0.6, 1.2, 0.8]
# sum(x * pmf) = 0.1 + 0.6 + 0.4 + 0.8 = 2.7
print(mean_value)
```

```
## [1] 2.7
```

To compute $Var(X)$ we need to compute first:

$$E[X^2] = \sum_{x=1}^{4} x^2 P(X = x) = 1^2 \cdot 0.1 + 2^2 \cdot 0.3 + 3^2 \cdot 0.4 + 4^2 \cdot 0.2 = 8.1$$

```
x <- 1:4
pmf <- c(0.1, 0.3, 0.4, 0.2) # Creates a column, i.e., a vector.
mean_sq_value <- sum(x^2 * pmf)
print(mean_value)
```

```
## [1] 2.7
```

So:

$$Var(X) = 8.1 - 2.7^2 = 8.1 - 7.29 = 0.81.$$

To verify using R:

```r
var_x <- mean_sq_value - mean_value^2
print(var_x)
```

```
## [1] 0.81
```

> Observe that Var(x) is always non-negative, so, in case, e.g., of an exam, negative values are not... appreciated.

To compute the standard deviation:

$$SD(X) = \sqrt{Var(X)} = \sqrt{0.81} = 0.9.$$

> The striking difference between SD and Var are the units in which each is measured. Var(X) is measured in square units (whatever the unit of X) while SD(X) is measured in the same units as X.

Exercise 2

In a children's board game, players spin a pointer on a wheel. The wheel determines how many spaces a player moves forward or backward. Let Y be the number of spaces moved.

- $Y = -1$ (Move backward 1 space) with probability 0.20
- $Y = 0$ (Stay put) with probability 0.15
- $Y = 2$ (Move forward 2 spaces) with probability 0.45
- $Y = 5$ (Move forward 5 spaces) with probability 0.20

Questions:

a) Construct a formal probability distribution table for Y.
b) What is the average (expected) number of spaces a player moves per spin?
c) Calculate the variance and standard deviation of Y to measure the variability of the spins.

Solution The PMF table is shown below:

y	$P(Y = y)$
-1	0.20
0	0.15
2	0.45
5	0.20

> In general, the *average* or *expected* value of something is its mean value, i.e., $E[X]$.

Using the formula:

$$E[X] = \sum_y yP(Y = y) = 1.7.$$

We can also compute $Var(Y)$ and $SD(Y)$ using R:

```r
y <- c(-1, 0, 2, 5)
pmf <- c(0.20, 0.15, 0.45, 0.20)
var_y <- sum(y^2 * pmf) - (sum(y * pmf)) ^ 2
sd_y <- sqrt(var_y)
print(paste('Variance:', var_y, 'Standard Deviation:', sd_y))
```

```
## [1] "Variance: 4.11 Standard Deviation: 2.02731349327133"
```

Exercise 3

A small tech support center tracks the number of urgent tickets, Z, they receive in the first hour of a shift. The distribution is partially given below:

z	$P(Z = z)$
0	0.40
1	0.35
2	k
3	0.05

Questions:

a) Find the missing value k that makes this a legitimate probability distribution.
b) Find the mean number of urgent tickets received during this hour.
c) Find the standard deviation of the tickets.

Solution To compute the missing value, we use the fact that all probabilities must sum to 1:

$$0.4 + 0.35 + k + 0.05 = 1 \iff 0.80 + k = 1 \iff k = 0.20.$$

We compute the rest using R:

```r
z <- 0:3
pmf <- c(0.40, 0.35, 0.20, 0.05)
mv <- sum(z * pmf)
varz <- sum(z^2 * pmf) - (sum(z * pmf))^2
sdz <- sqrt(varz)
print(paste('Mean:', mv, 'Variance:', varz, 'Standard Deviation:', sdz))
```

```
## [1] "Mean: 0.9 Variance: 0.79 Standard Deviation: 0.888819441731559"
```

Exercise 4

For each of the following scenarios, identify the appropriate probability distribution from Part 1 (Bernoulli, Binomial, Poisson, Geometric, or Negative Binomial) and use the formulas from the table to state the **mean** and **standard deviation**.

1. A basketball player has an 80% free-throw success rate. Let A be the number of free throws made out of a total of 15 independent attempts.
2. A router receives an average of 4.5 web traffic requests per millisecond. Let B be the total number of requests received in a randomly selected millisecond.
3. A quality assurance inspector tests computer chips one by one until they find their very first defective chip. The probability of any chip being defective is 0.02. Let C be the total number of chips tested.

Solution Since each free-through is either in or out, we have $n = 15$ repeated Bernoulli trials with $p = 0.8$, so A follows a Binomial distribution $(n, p) = (15, 0.8)$. So:

$$E[A] = np = 12, \quad SD(A) = \sqrt{np(1-p)} = \sqrt{2.4} \approx 1.55.$$

Since we know the **average rate** of request per some unit, B follows a Poisson distribution with $\lambda = 4.5$. So:

$$E[B] = \lambda = 0.5, \quad SD(B) = \sqrt{\lambda} = \sqrt{0.5} \approx 0.71.$$

For C, since we count *trials* until fist success (spotting a failing chip), C follows a geometric distribution:

$$E[C] = \frac{1}{p} = 50, \quad SD(C) = \frac{\sqrt{1-p}}{p} = \frac{\sqrt{0.98}}{0.02} = 49.5.$$

Problems

Problem 1

An independent contractor reviews insurance claims. The number of claims they process in a single day, X, is a random variable with mean $E[X] = 6.5$ and variance $Var(X) = 2.25$.

The contractor is paid a fixed daily base rate of \$150 plus an additional \$40 per claim processed. Let W represent the contractor's total daily earnings ($W = 40X + 150$).

Questions:

a) Calculate the contractor's expected daily earnings, $E[W]$.
b) Calculate the variance of their daily earnings, $Var(W)$.
c) Find the standard deviation of their daily earnings, $SD(W)$. *Hint: Recall how constants affect mean and variance:* $E[aX + b] = aE[X] + b$ *and* $Var(aX + b) = a^2 Var(X)$.

Solution We can compute the mean value directly using the definition:

$$
\begin{aligned}
E[W] = \sum_{w} wP(W = w) = \sum_{x=0}^{\infty}(40x + 150)P(X = x) = \\
= \sum_{x=0}^{\infty} 40xP(X = x) + \sum_{x=0}^{\infty} 150P(X = x) = \\
= 40\sum_{x=0}^{\infty} xP(X = x) + 150\sum_{x=0}^{\infty} P(X = x) = \\
= 40E[X] + 150 \cdot 1 = \\
= 40 \cdot 6.5 + 150 = \\
= 410
\end{aligned}
$$

Or, we can just use the linearity of mean values:

$$E[aX + b] = aE[X] + b.$$

So, in our case:

$$E[W] = E[40X + 150] = 40E[X] + 150 = 410.$$

Similarly, for the variance, we know that:

$$
\begin{aligned}
Var(aX) = a^2 Var(X) \\
Var(X + b) = Var(X)
\end{aligned}
$$

So, in our case:

$$Var(W) = Var(40X + 150) = 40^2 Var(X) = 3600.$$

For the standard deviation:

$$SD(W) = \sqrt{Var(W)} = \sqrt{3600} = 60.$$

Problem 2

Two distinct software scripts run independently on a server. Let X be the number of errors script A encounters per hour, and let Y be the number of errors script B encounters per hour. The joint probability distribution $P(X = x, Y = y)$ is given below:

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.12	0.18	0.10
$X = 1$	0.18	0.27	0.15

Questions:

a) Find the marginal probability distributions for both X and Y.
b) Are X and Y independent random variables? Prove your answer mathematically.
c) Calculate $E[X]$, $E[Y]$, and the expected value of their product, $E[XY]$.

Solution To compute the marginal probability distributions we can sum over rows (for X) and columns (for Y):

	$Y=0$	$Y=1$	$Y=2$	Total
$X=0$	0.12	0.18	0.10	**0.40**
$X=1$	0.18	0.27	0.15	**0.60**
Total	**0.30**	**0.45**	**0.25**	1.00

To verify that two variables are independent we must verify whether the following holds:

$$P(X=x, Y=y) = P(X=x)P(Y=y).$$

So, to verify this, we can leverage some linear algebra:

```
x <- c(0.30, 0.45, 0.25)
y <- c(0.40, 0.60)
xy <- y %*% t(x)
print(xy)
```

```
##      [,1] [,2] [,3]
## [1,] 0.12 0.18 0.10
## [2,] 0.18 0.27 0.15
```

In the above, we have taken advantage of matrix multiplication, since:

$$\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} \begin{pmatrix} 4 & 2 \end{pmatrix} = \begin{pmatrix} 2\cdot 4 & 2\cdot 2 \\ 1\cdot 4 & 1\cdot 2 \\ 5\cdot 4 & 5\cdot 2 \end{pmatrix} = \begin{pmatrix} 8 & 4 \\ 4 & 2 \\ 20 & 10 \end{pmatrix}$$

So, essentially, we multiply each row of the left matrix (column) with each column of the right matrix (row). Thus, we end up with all possible products.

But, since we are not required to know anything about Linear Algebra, a more formal proof would be to just compute all pairwise products and cross check them with the original distribution table:

	$Y=0$	$Y=1$	$Y=2$	Total
$X=0$	0.40*0.30=0.12	0.40*0.45=0.18	0.40*0.25=0.10	**0.40**
$X=1$	0.60*0.30=0.18	0.60*0.45=0.27	0.60*0.25=0.15	**0.60**
Total	**0.30**	**0.45**	**0.25**	1.00

So, in any case, the two variables are independent!

For the corresponding mean values:

$$
\begin{aligned}
E[X] &= 0 \cdot 0.40 + 1 \cdot 0.60 = 0.60 \\
E[Y] &= 0 \cdot 0.30 + 1 \cdot 0.45 + 2 \cdot 0.25 = 0.95 \\
E[XY] &= \sum_{x,y} xyP(X = x, Y = y) = \\
&= \sum_{x} \sum_{y} xyP(X = x)P(Y = y) = \\
&= \sum_{x} xP(X = x) \sum_{y} yP(Y = y) = \\
&= E[X]E[Y] = \\
&= 0.60 \cdot 0.95 = \\
&= 0.57.
\end{aligned}
$$

Attention! The identity $E[XY] = E[X]E[Y]$ is true if X, Y are pairwise independent!

Problem 3

Use the formulas from the reference table in Part 1 to solve the following applied problems:

1. A customer support line receives an average of $\lambda = 12$ calls per hour.
   a) What is the expected number of calls received over a 3-hour period?
   b) What is the standard deviation of the number of calls received over that same 3-hour period?
2. A factory production line creates mechanical parts, where each part has a 5% chance ($p = 0.05$) of being defective. The supervisor runs the assembly line until they accumulate exactly $r = 3$ defective parts for quality testing.
   a) What is the expected total number of parts the line must produce before stopping?
   b) Calculate the standard deviation of the total number of parts produced.

Solution Since we know the (constant) average call rate, letting X be the number of calls over a 3-hour period will also follow a Poisson distribution with average rate of:

$$\lambda' = 3\lambda = 36.$$

So, the expected number of calls will be:

$$E[X] = \lambda' = 36.$$

Similarly:

$$SD(X) = \sqrt{36} = 6.$$

Let now Y be the number of produced parts until $r = 3$ defective parts are produced. Then, Y follows a negative binomial distribution with parameters $(r, p) = (3, 0.05)$, so:

$$E[Y] = \frac{r}{p} = \frac{3}{0.05} = 60.$$

Similarly:

$$SD(Y) = \frac{\sqrt{r(1-p)}}{p} = \frac{\sqrt{3 \cdot 0.95}}{0.05} \approx 33.76.$$